

Date: 23/07/2025

Lab Practical #8:

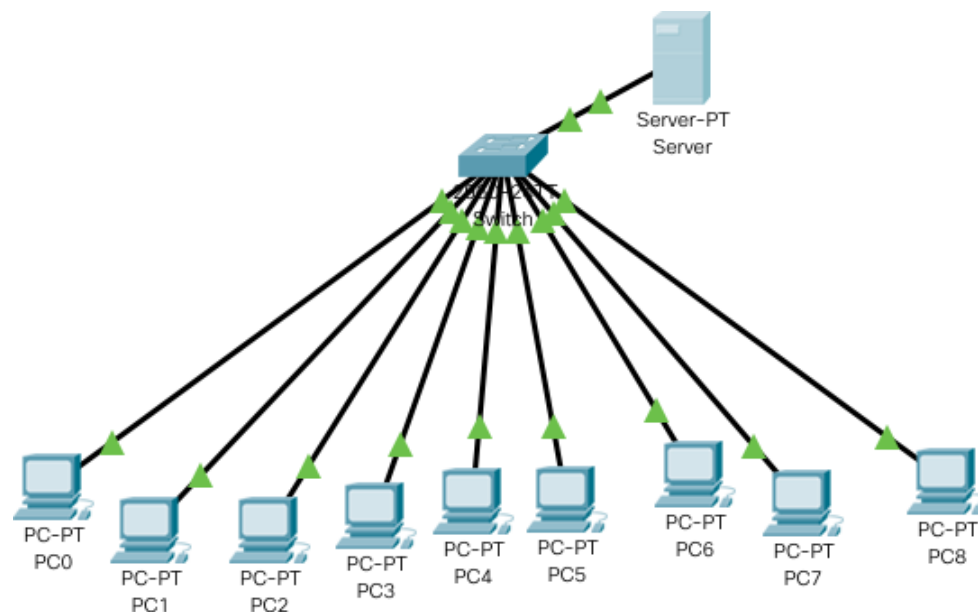
Study & Survey of Institute organization network infrastructure.

Practical Assignment #8:

1. Identify type of network in your institute. Draw a design of network in your institute (Any Lab/Floor/Building).

Lab No: C- 306

Topology: Star Topology



2. List how many network devices and types of cable used and give its details.

Networking Devices :

1. Switch

- Connects many devices in the same network, like PCs and printers.
- Sends data only to the device it is meant for, not to everyone.

2. Router

- Joins two or more networks together, like your home network and the internet.
- Chooses the best path for data to travel from one place to another.

3. Firewall

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- the network by blocking unwanted or unsafe traffic.
- Works based on rules set by the network administrator.

4. Network Interface Card (NIC)

- A hardware component that allows a computer or device to connect to a network.
- Can be wired (Ethernet) or wireless (Wi-Fi).

5. Access Point (AP)

- Wi-Fi so that wireless devices can join the network.
- back to a wired network to give internet access.

6. Load Balancer

- the network traffic across different servers.
- Helps improve speed, availability, and reliability of services.

7. Modem

- digital signals to analog (and back) so devices can access the internet.
- Works with telephone lines, cable lines, or fiber.

Types of Network Cables:

1. Fiber Optic Cable

- Transfers data using light signals, which makes it very fast.
- Can send information over long distances without losing quality.

2. Ethernet Cable (Cat5e, Cat6, Cat6a)

- Most common cable used for wired connections.
- Made of twisted pairs that reduce noise and support high speed.

3. Console Cable (RJ45 to Serial)

- Special cable used for configuring routers and switches.
- Connects a network device directly to a computer.

4. Coaxial Cable

- Older type of cable that carries data with electrical signals.
- Still used in some TV and internet connections.